$$\mathbb{E}\left(\frac{\partial l}{\partial \sigma}\right)^2 = \mathbb{E}\left[\left(-\frac{n}{\sigma} + \frac{\|X-\mu\|^2}{\sigma^3}\right)^2\right]$$

$$= \mathbb{E}\left[\frac{n^2}{\sigma^2} + \frac{\left(\|X-\mu\|^2\right)^2}{\sigma^6} - \frac{2n\|X-\mu\|^2}{\sigma^4}\right]$$

$$= \cancel{\mathbb{E}}\,\frac{n^2}{\sigma^2} + \frac{\mathbb{E}\left[\left(\sum_{i=1}^{n}(X_i-\mu_i)^2\right)^2\right]}{\sigma^6} - \frac{2n\,\mathbb{E}\left[\sum_{i=1}^{n}(X_i-\mu_i)^2\right]}{\sigma^4}$$

$$= \frac{n^2}{\sigma^2} + \frac{\mathbb{E}\left[\sum_{i=1}^{n}(X_i-\mu_i)^4 + 2\sum_{1\le i<j\le n}(X_i-\mu_i)^2(X_j-\mu_j)^2\right]}{\sigma^6} - \frac{2n\sum_{i=1}^{n}\mathbb{E}\left[(X_i-\mu_i)^2\right]}{\sigma^4}$$

$$= \frac{n^2}{\sigma^2} + \frac{\sum_{i=1}^{n}\mathbb{E}\left[(X_i-\mu_i)^4\right] + 2\sum_{1\le i<j\le n}\mathbb{E}\left[(X_i-\mu_i)^2\right]\cdot\mathbb{E}\left[(X_j-\mu_j)^2\right]}{\sigma^6} - \frac{2n^2\sigma^2}{\sigma^4}$$

$$= \frac{-n^2}{\sigma^2} + \frac{n\cdot 3\sigma^2 + 2\cdot\frac{n(n-1)}{2}\cdot\sigma^4}{\sigma^6}$$

$$= \frac{1}{\sigma^2}\left(-n^2 + 3n + n^2 - n\right)$$

$$= \frac{2n}{\sigma^2}$$